COURSE
OVERVIEW

TELECOMS TOWER ABSEILING

The Telecommunication Tower Abseiling course is designed to provide learners with the knowledge and practical skills required to access and work safely on telecommunication towers using controlled abseiling techniques.

The course focuses on work positioning, rope systems, anchor selection, and the application of fall protection principles where conventional access methods are not feasible. Emphasis is placed on hazard identification, maintaining two points of attachment, equipment selection, and safe descent and positioning to reduce the risk of falls and injury.

BY COMPLETING THIS COURSE, LEARNERS WILL BE ABLE TO:

- Identify hazards associated with telecommunication tower abseiling activities
- Apply correct work positioning and abseiling techniques
- Select, inspect, and use abseiling and fall protection equipment correctly
- Maintain two points of attachment at all times
- Identify suitable anchor points for tower abseiling operations
- Follow safe work procedures when accessing and working on towers

INDEX

INTRODUCTION

GRAVITY VALUES

R — RELATIONSHIPS

Creating and maintaining meaningful relationships with all stakeholders of Gravity.

I — INTEGRITY

Conducting day-to-day business with integrity at all times.

S — SOLUTIONS **DRIVEN**

Being solution driven.

Q — QUALITY & **EXCELLENCE**

Striving for excellence and constant improvement.

WELCOME, LEARNER!

Working at height is inherently risky — and this includes training exercises. The best way to stay safe is by learning and practising the correct techniques. This manual will help you:

- Understand the basics of fall arrest and basic rescue
- Learn how to identify and manage height-related hazards
- Build the skills needed to work safely and confidently

Working at height means doing any job where you could fall from one level to another. This includes tasks done:

- Above the ground or floor
- Near open edges
- Near fragile surfaces
- Next to holes or gaps in the ground or floor
- Close to excavations

Note: Work at height does not include slipping or tripping hazards on the same level.

HOW TO USE THIS MANUAL?

This guide supports the Fall Arrest and Basic Rescue Course and is meant to be used with the help of a trained instructor.

At Gravity Training, all facilitators are highly trained to deliver top-quality learning experiences. The content has been designed to be clear, easy to remember, and practical for real worksites.

SITE ESTABLISHMENT

Before any work begins, the site must be safe.
Use this simple checklist

01 Follow the Rules:

Make sure you follow all client rules.

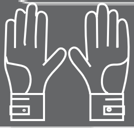
Example: If night work is not allowed, do not work at night.

02 Rescue Plan

Have a rescue plan ready. A rescue kit and first aid kit must be on site.

03 Work in Pairs

There must be at least two workers on site.
If something goes wrong, one can help or call for help.



04 Check Equipment

All tools and gear must be safe and working before use.

05 Safe Site

Check the building or structure.

Make sure it is strong and safe to work on.

06

Manage Traffic

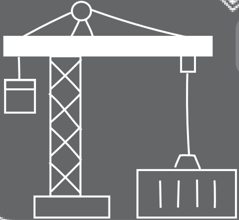
If there are cars or trucks nearby, there must be a traffic safety plan.

07

Do a Risk Assessment

Check what dangers there are.

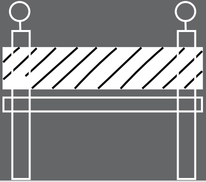
All workers must understand the risks before starting work.



08

Use the Right Safety Gear

Make sure the fall protection fits the job and the site.



EXCLUSION ZONES **& BARRICADES**

Exclusion Zone: An area where no one is allowed to walk or work.

Example: A 2-metre space around a hole in the roof.

Barricade: A physical barrier to block people from danger.

Example: A fence or tape to stop people from falling.

SITE SAFETY

Be healthy and ready before starting work.

Only do jobs you are trained and allowed to do.

Never work under the influence of alcohol or drugs.

Always wear your safety gear (PPE, harness if working at height).

Use harnesses correctly and stay connected at heights.

Never work alone and always follow safety steps.

SITE SAFETY

Keep people away from areas below when working above.

Don’t walk under lifted loads or stand in dangerous areas.

Only trained people can do electrical, underground, or confined space work.

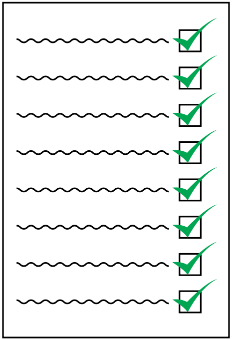
Follow road rules. Wear helmets. No passengers on open trucks..

Don’t drive tired or distracted.

Stop unsafe work! Everyone has the right to speak up.

1. Risk Assessment

The most important part of the Fall Protection Plan is the **Risk Assessment.**



What to do when approaching a work site

STEP 1 : RISK IDENTIFICATION

The following are factors must be considered when identifying risks:

- **Site requirements**
- **Access method requirements**
- **Required team composition**
- **Fall protection requirements**
- **PPE requirements**
- **Environmental**

After this, the risks must be communicated to everyone on site by means of:

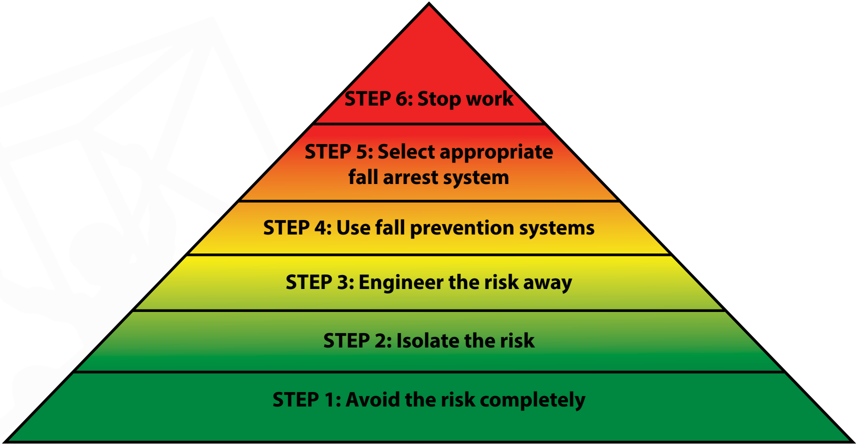
- Site induction
- Toolbox talk

STEP 2 : PRE-MITIGATION RISK RATING

LIKELIHOOD OF OCCURRENCE	HAZARD SEVERITY				
	Negligible	Slight	Moderate	High	Very High
Very Unlikely	LOW	LOW	LOW	LOW	LOW
Unlikely	LOW	LOW	LOW	MEDIUM	MEDIUM
Possible	LOW	LOW	MEDIUM	MEDIUM	HIGH
Likely	LOW	MEDIUM	MEDIUM	HIGH	HIGH
Very Likely	MEDIUM	MEDIUM	HIGH	HIGH	HIGH

No work is allowed to commence if there is a "Medium" or "High" risk present on site.
Thus, we have to mitigate all identified risks in order for them to receive a "Low" risk rating.

STEP 3 : RISK MITIGATION HIERARCHY



- **Adverse Weather Conditions:**
  No work-at-height shall be conducted during any inclement weather conditions. Should the situation change whilst at height, all technicians must safely descend to the ground and the Site Supervisor must report back immediately to the Project Manager.

  Examples of inclement weather conditions:
  - Rain
  - Lightning
  - Extremely cold weather
  - Extremely hot weather
  - Sleet or snow
  - High or gusty wind
  - Poor visibility (fog or smog)

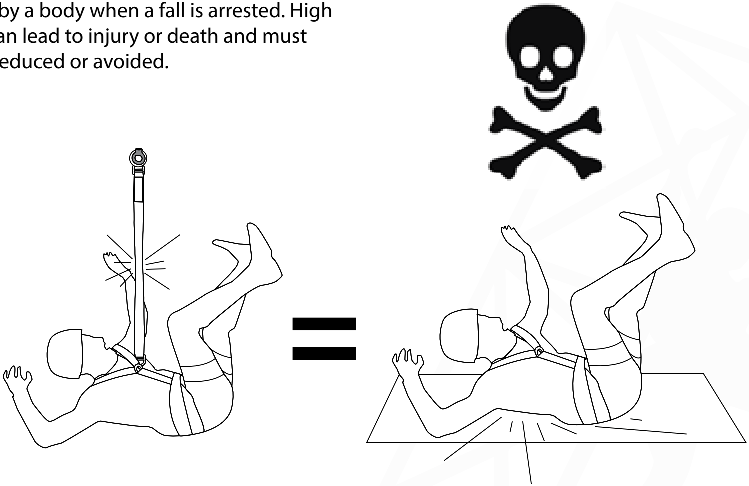
STEP 4 : POST-MITIGATION RISK RATING

		HAZARD SEVERITY				
		Negligible	Slight	Moderate	High	Very High
LIKELIHOOD OF OCCURRENCE	Very Unlikely	*LOW*	*LOW*	*LOW*	*LOW*	*LOW*
	~~Unlikely~~	~~*LOW*~~	~~*LOW*~~	~~*LOW*~~	~~*MEDIUM*~~	*MEDIUM*
	Possible	*LOW*	*LOW*	*MEDIUM*	*MEDIUM*	*HIGH*
	Likely	*LOW*	*MEDIUM*	*MEDIUM*	*HIGH*	*HIGH*
	Very Likely	*MEDIUM*	*MEDIUM*	*HIGH*	*HIGH*	*HIGH*

No work is allowed to commence if there is a "**Medium**" or "**High**" risk present on site.
Thus, we have to mitigate all identified risks in order for them to receive a "**Low**" risk rating.

B. Work at Height Basics
1. Shock Load

The force felt by a body when a fall is arrested. High shock loads can lead to injury or death and must therefore be reduced or avoided.



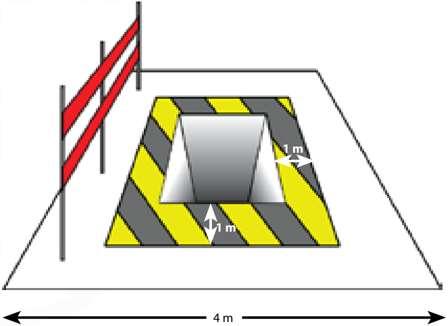
C. Site Establishment.

1. Before Work Starts

- Ensure that you comply with the client requirements on site. E.g. night work is allowed etc.
- Before work starts, ensure that there are at least 2 workers on site with the ability to rescue each other.
- Ensure that the fall protection system is relevant and fit for purpose as per site and task.
- Ensure that a comprehensive risk assessment has been done and that all workers have been inducted on the risk assessment.
- Ensure that all equipment is inspected before use and safe for use and fit for purpose.
- Ensure that a rescue plan is place and a sufficient rescue kit is on site with a first aid kit.

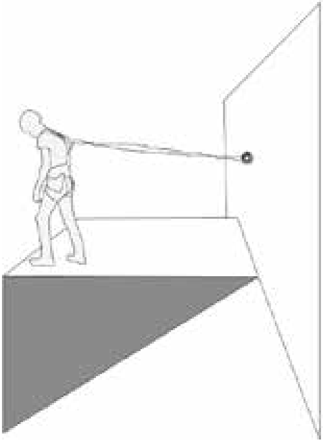
2. Exclusion Zones and Barricading

- **Exclusion zones:** A zone or space where people are **not allowed** to walk or work within, for example, a 1 m space around a hole in a roof.
- **Barricading**: A physical barrier **preventing** people from reaching a fall risk position.



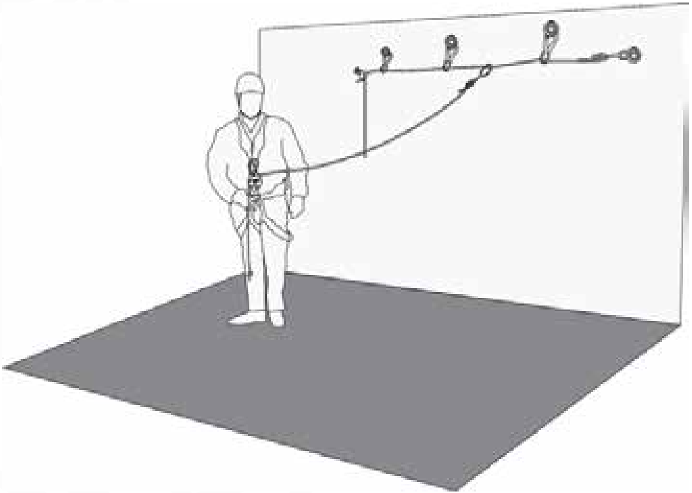
3. Work Restraint

A system where we connect a worker to the structure in such a way that it prevents him/her from reaching the edge, thus preventing a fall.



4. Roof Top Work Restraint System

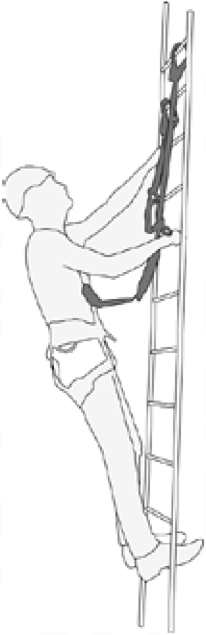
One can combine the horizontal lifeline with a work restraint system to create a system ideal for rooftop work.



5. Fall Arrest

A system used to suspend a person safely AFTER a fall has occurred, and doing all this in such a way as to prevent injury as far as possible.

Fall Arrest does NOT prevent a fall like Work Restraint. It only prevents a person from serious injury or hitting the ground due to a fall by reducing the shock load.



6. Telecommunication Tower Abseil

Definition: A form of work positioning, using a series of connection points and ropes to enable access to difficult to reach locations where other safer methods are not possible.

Much like Fall Arrest, Rope Access does NOT prevent a fall like Work Restraint does. It only prevents a person from seriously getting hurt or hitting the ground due to a fall. Rope Access also employs a double rope system that reduces the likelihood of a ground fall greatly.

2-Point Principle

A rope access technician must at all times have two points of connection, i.e. one point on each of the two ropes. Failure to do so will put the technician in grave danger of a ground fall.

Please note, although telecommunication tower abseiling is similar to rope access, it is not to be confused with rope access as not all the techniques, maneuvers and procedures used in rope access applies to telecommunication tower abseiling.

6. Full Range of Personal Protective Equipment (PPE)

In order for a user to use PPE safely and correctly, there are four things each user must know about each piece of equipment, namely:

- **Identification** - The name and purpose of each piece of equipment.
- **Certification** - Only equipment with the correct South African National Standards (SANS) or European National Standards (EN) should be used and must be purchased through a reputable retailer such as Gravity Training. For example, Gravity Training only makes use of ISO 9001:2015 certified equipment suppliers.
- **Limitation** - How to use and not use equipment, the safe use criteria and hazards to identify when using each piece of equipment, and the maximum load that each piece of equipment is certified to carry. Equipment load bearing ability is given by the MANUFACTURER as **Safe Working Load** and the **Breaking Strength** and should never be exceeded.
- **Inspection** - Pre-use inspection should be done before start of work and that equipment should be sent to the appointed competent equipment controller for detailed inspections at least every 6 months. The use of faulty or damaged equipment may lead to serious injury or death.

Climbing Helmet (EN 12492 :2012)

A helmet is used to protect the technician's head during a fall as well as from falling objects:

- Must have chinstrap
- Must be on worker's head at all times
- Chinstrap is designed to break under force.

Full Body Harness (EN361:2002, 358:2000, EN 813:2008)

A harness allows all other devices and systems to be connected to the worker. This forms the basis of all safety systems.

- Fall arrest systems must preferably be connected the chest D-link.
- The two buckles on the side of the harness are for work positioning systems.
- The harness must be fitted correctly with all straps tightened properly.
- Ensure that the buckles go to the correct corresponding connectors.

- S.W.L. = 150kg
- B.S. = 1500kg

Donning and duffing a harness

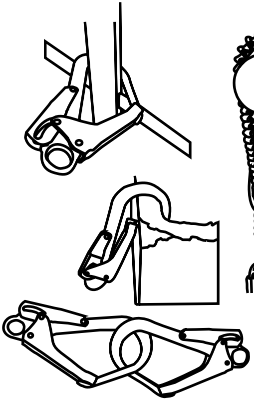
When donning a harness ensure the following:
The waistbelt is above your hips and tight.
The should straps are tight so that you can fit a fist between your chest and the harness.
The leg loops are adjusted so that they do not restrict blood flow when you crouch.
All the buckles are connected and secure.

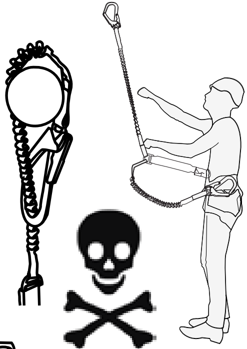
Shock Absorbing Lanyard (EN 355:2002)

A shock absorber acts as the Primary safety when working at height.

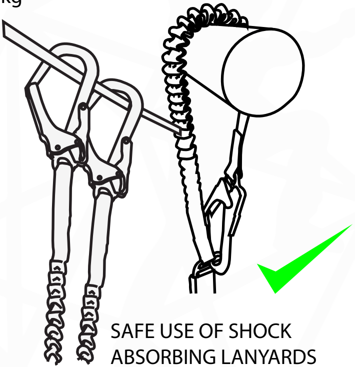
- Shock absorber rips open (deploys) to reduce the shock load felt by the person falling.
- Must be intact during use. Once activated, the shock absorber may not be used again as it will not be able to absorb the shock load of another fall.

- S.W.L. = 150kg
- B.S. = 1500kg



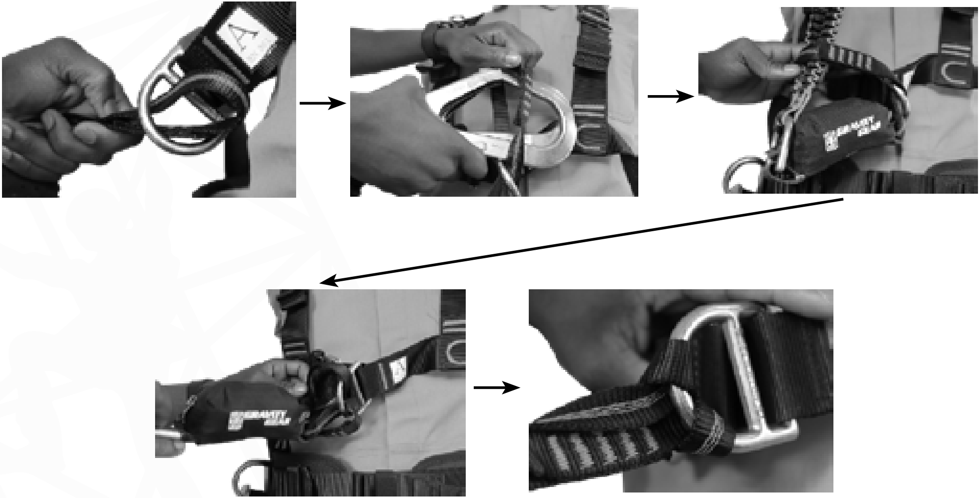


UNSAFE USE OF SHOCK ABSORBING LANYARDS



SAFE USE OF SHOCK ABSORBING LANYARDS

How to fit the Gravity Shock Absorbing Lanyard

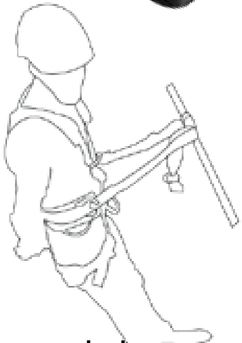


Work Positioning Lanyard (EN 358:2000)

A work positioning lanyard is used to allow a worker to use both hands freely in order to do work effectively.

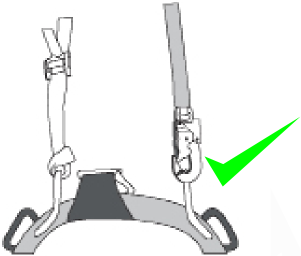
- MUST NEVER BE USED WITHOUT A FALL ARREST SYSTEM.
- Should be wrapped around the structure to prevent slipping.

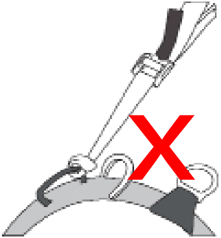
- S.W.L. = 150kg
- B.S. = 1500kg

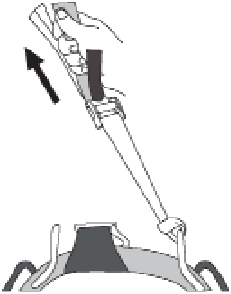


The helmet, harness, shock absorber and work positioning lanyard form the minimum PPE that must be used by a worker when at height.

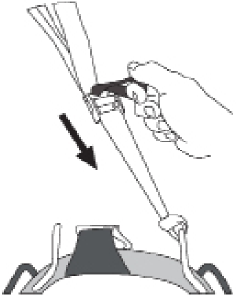
How to fit the Gravity Work Positioning Lanyard







How to adjust the Gravity Work Positioning Lanyard



Rope Grabs Devices (EN 353-2:2002)

Used on a vertical lifeline as a Fall Arrest device. This will grab the rope in the event of a fall, limiting the falling distance to only rope stretch.

- Must be used in a safe fall factor.
- Used on a vertical lifeline.
- Must not be manipulated directly as this may interfere with the working action of the device.

- S.W.L. = 150kg
- B.S. = 1500kg

Goblin Rope Surfer (EN 567:2017 & EN 795:2012)

Used to connect the backup device to the harness in order for the backup device to be able to move freely in a Fall Factor 0.

- Connects to the top D-Link.
- Stitched end goes to the harness.

- **Safe Working Load : 150 kg**
- **Breaking Strength : 1500 kg**

Descending Devices (EN 341:2011)

A variable friction device that controls the rate of descent. Used for an emergency escape, horizontal lifelines and rescue.

- Must be locked when not in use.
- Used during emergency escape, when tensioning a horizontal lifeline and during rescue.
- When using in rescue, extra friction must be applied.

- S.W.L. = 150kg
- B.S. = 1500kg
- S.W.L (with extra friction) = 250kg

Slings (EN 566:2017 & 795:2012)

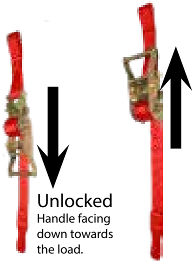
Used to anchor systems to structures when sharp edges are not present.

- Should be wrapped around the structure to prevent slipping.
- Stitching must be straight to prevent the weakening of the sling.
- Slings with PVC sleeves should be used should sharp edges be a concern.

- S.W.L. = 220kg
- B.S. = 2200kg

Rescue Ratchet Slings (SANS 50354:2003)

Locked
Handle facing upwards towards the anchor.



Unlocked
Handle facing down towards the load.

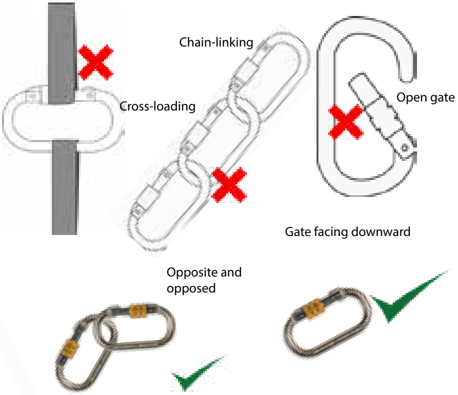
Used during the ratchet rescue to lift casualty up to allow rescuer to remove the casualty's lanyards.

- Has a ratchet mechanism to allow for easy lifting.
- Must be locked (closed completely) when not in use.
- Always reset the ratchet after use.

- S.W.L. = 150kg
- B.S. = 1500kg

Connectors (EN 362:2004)

Used to connect a range of devices and different pieces of equipment.



- S.W.L. = 220kg
- B.S. = 2200kg

Work Ropes
Semi-Static Ropes (EN 1891:1998)

Ribbon stating date of manufacture

Kern\Core

Sheath\Mantle

Used throughout work at height systems and is a crucial part of height safety.

- Stretches between 5% and 10%.
- Is of kermantle construction. Consists of two parts:
  Kern (Core) = 80% of the strength.
  Mantle (sheath) = 20% of the strength.

- S.W.L. = 250kg
- B.S. = 2500kg

Cowtails (EN 892:2016)

Cowtails refer to three separate lengths of dynamic rope which are connected to the harness and allows the worker to connect a range of devices onto him/herself.

- Dynamic rope stretches between 20% and 25%.
- Connected with a Figure 8 Follow Through to the harness.
- Make a Figure 8 on a Bight on the other end.
- Ensure all knots are dressed correctly and neatly.
- Ensure that the cowtails are the correct length for the task.

- **Safe Working Load : 250 kg**
- **Breaking Strength : 2500 kg**

Ascenders (EN 567:2013)

Ascenders refer to any device that allows a worker to climb UP on a rope. They all have a cam with downward facing teeth that bite into the rope, thus preventing them from moving downward. They are divided into two groups, namely:

Handled Ascender (Jumar)

- Must always be closed - on and off the rope.
- Never shock load an ascender as this can cause rope damage.

- **Safe Working Load : 100 kg**
- **Breaking Strength : 1000 kg**
- **Rope Damage : 400 kg**

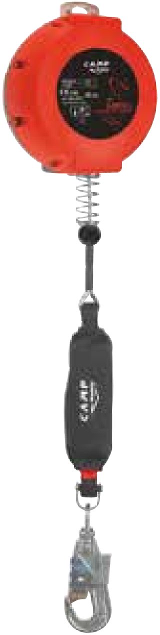
Etrier

An etrier, or “step-loop”, is used alongside the handled ascender in order for the worker to be able to move the chest ascender upwards in order to move up a rope.

Etriers are not load bearing, thus they do not have ratings or weight specifications.

However, it is important that they remain in a good condition; if they were to break while at height, one will have a difficult time removing one’s chest ascender.

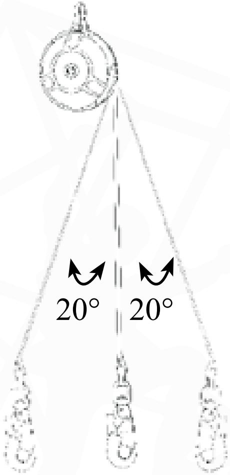
Retractable Lanyard (EN 360)



Used as a self-retracting fall arrest device, this allows quick and easy movement up and down. The device limits falling distance to approximately 30 cm.

- Similar functionality of a car safety belt; it allows a technician to freely move up and down, and locks only when a person has fallen.
- Can be used for work positioning and fall arrest.
- Some models include a shock-absorbing function.
- Has a 20 degree maximum range during vertical use.
- Never use above a Fall Factor 1.

- S.W.L. = 120kg
- Please note the S.W.L might differ from manufacturer or by design.



Pulleys (EN 12278:2007)

Pulleys preform two main functions:

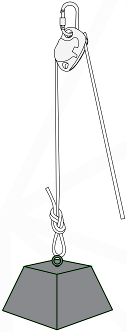
1. Reduce friction in a system.
2. Change the direction in which a worker needs to pull in order to lift a load.

- S.W.L. = 160kg
- B.S. = 3200kg

Whatever force is applied to one side of the pulley is experienced on the other side, thus, the point of connection feels twice the weight of what is being lifted.

Carrying Tools at Height

All tools must be tied off with tool lanyards or accessory cord to prevent tools from being dropped from height. Tools being dropped can cause damage to property and even injury or death to persons below.



Tools weighing 8 kg or less should be connected to the harness gear loops. Tools weighing in excess of 8 kg should be lifted using a separate 1:1 pulley system with a maximum lifting capacity of 20 kg. Any tools or equipment weighing more than 20 kg need to lifted by a qualified rope rigger.

Principles of Slinging:

- Never exceed the safe working load of equipment.
- Always protect slings from sharp edges.
- Always perform a pre-lift before lifting.
- Always consider the center of gravity.

Principles of Rigging:

- Never stand underneath a load.
- Always ensure that the route traveled is clear of obstructions.
- Ensure that the correct equipment is used for the lift.

Pre-use Inspections

Users must inspect all the equipment to be used before work starts. This must be done to ensure that all equipment is safe for use and fit for purpose in accordance with ISO 22846-2.

Equipment is divided into 2 categories namely hardware and software, each of these have their own inspection criteria. Some pieces of equipment (e.g a harness) fall into both categories and must be inspected as such.

Software

Check for:

- Date of manufacture
- Condition of stitching
- Discolouration
- Cuts
- Burns
- Wear and tear
- Certification

Hardware

Check for:

- Rust
- Working action
- Movement
- Wear and tear
- Deformation
- Cracks

Equipment Storage

Equipment must be stored correctly and according to manufacturer's specifications to prevent damage to equipment when not in use.

Store Equipment:

- In a cool, dry place - This will help prevent mildew in software and rust in hardware.
- Free of chemicals - This will reduce the chance of chemical damage to equipment.
- Where there are no infestations (rats and moths etc.), which can damage software; ensure that there is a system in place to prevent this.
- In a secure location - This will prevent unauthorized personnel entering and potentially damaging equipment.

Rope Care

- Do not use rope for anything other than work at height.
- Do not run rope over sharp edges; use edge protection.
- Do not stand on or walk over a rope.
- Ensure that all devices are compatible with any devices to be used.

Rope Bagging:

It may be preferable to bag a rope instead of coiling it. In such cases, it is recommended to deviate the rope over the structure or ladder and pull the rope into the bag, this will prevent knotting and allows a rope to be bagged faster.

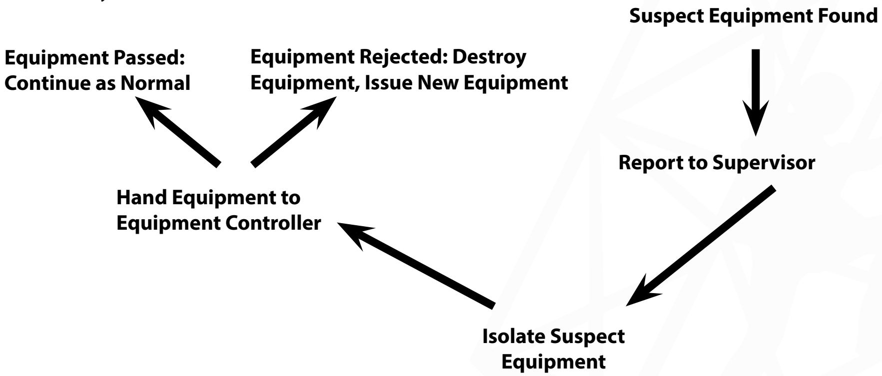
Housekeeping:

Whilst on site, it is important to ensure that the equipment being used is handled in such a way that they do not suffer any unnecessary damage or exposure to environmental conditions. When equipment is not in use , store them in their bags or bins. Ensure that equipment is moved out of the areas where workers move to prevent slips and trips or other accidents.

Reporting Suspect Equipment

Gear that is found to be damaged or too old for use must be reported to the person responsible for the site (either the Site Supervisor or Safety Officer etc).

Damaged or old equipment must be properly marked and removed from service immediately. Damaged equipment must be destroyed at the earliest opportunity to prevent the equipment from getting back into the work system.



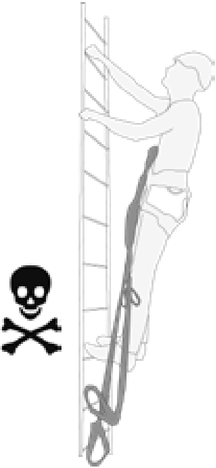
Repairing Damaged Equipment

Equipment may only be repaired by either the manufacturer or a company certified by the manufacturer.
NO CONTRACTOR MAY REPAIR PPE.

D. Work at Height Principles
1. Fall Arrest Fall Factors

Fall Factor =

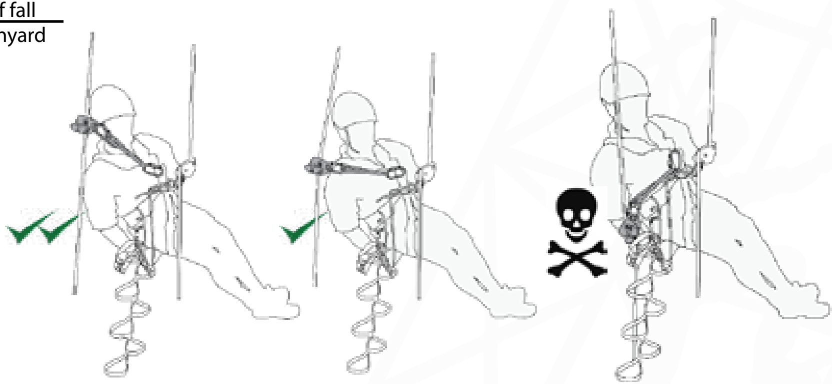
$$\frac{\text{Distance of fall}}{\text{Length of lanyard}}$$



Fall Factor 0: **Fall Factor 1:** **Fall Factor 2:**

2. Rope Access Fall Factors

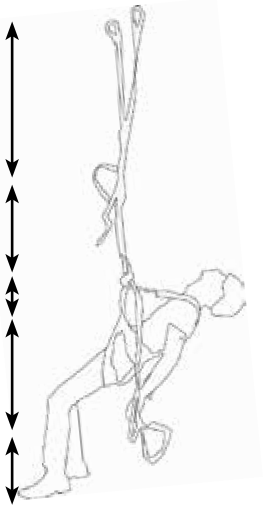
$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$



Fall Factor 0: **Fall Factor 1:** **Fall Factor 2:**

3. Minimum Free Space

Minimum free space (MFS) is the open space a technician would need below him to safely arrest a person's fall without hitting the ground or a structure. Reducing the fall distance or Fall Factor would reduce the required MFS.



2m for the lanyard

1.5m for the lanyard, since at Fall Factor 2 the lanyard will deploy completely

0.5m for the harness stretch

1.5m for the person's body.

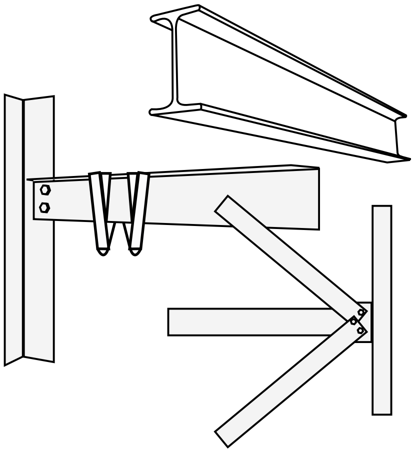
0.5m free space so that the person does not hit the ground

THUS: 2 m+1.5 m+0.5 m+1.5 m+0.5 m = 6 m

E. Anchor Points

Anchor points are mainly divided into two categories:

- Self-identified anchor points, and
- Planned anchor points (EN 795:2012).



Gravity Access Anchor

Eye-nut

Self-identified

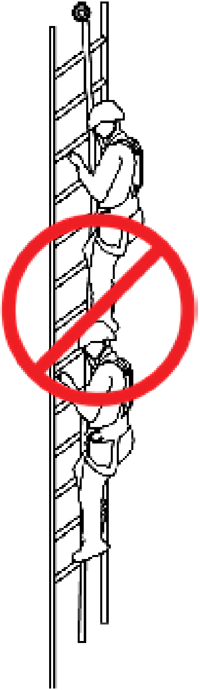
- S.W.L. = 150kg
- B.S. = 1500kg

Planned

- S.W.L. = 120kg
- B.S. = 1200kg

1. Temporary Vertical Lifelines

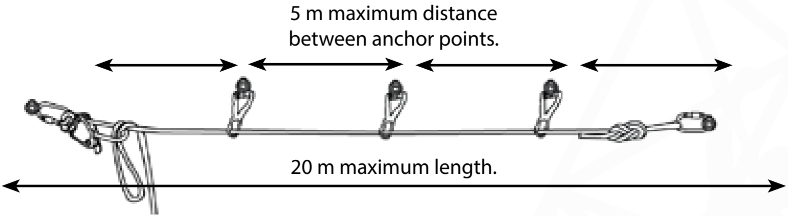
Vertical lifelines are used for easy and quick movement up and down a structure. These lifelines are used with a rope grab device which results in a reduced falling distance, only being limited to the amount of rope stretch.

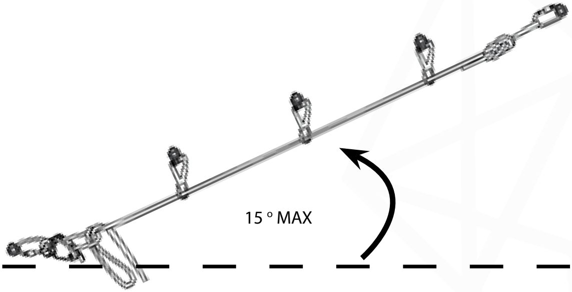


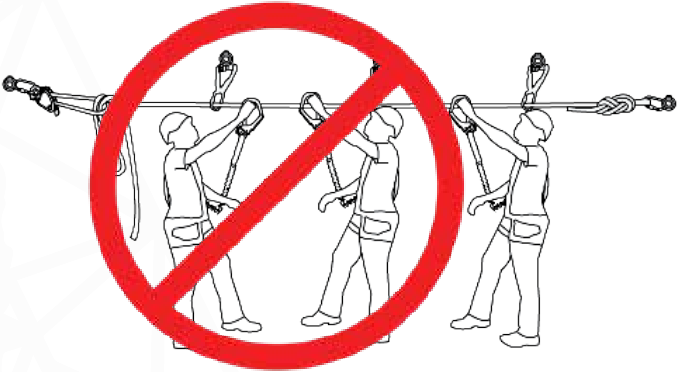
- Maximum Length: 30 m
- Maximum persons allowed on the lifeline: 1 Person.

2. Temporary Horizontal Lifelines

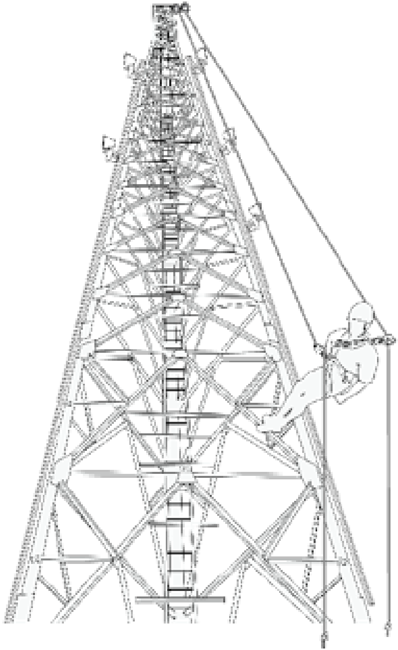
Horizontal lifelines are used for safe and quick movement from side to side. These lifelines are to be used with a shock absorber which can slide from side to side as the technician moves.







- No more than 2 users on a line
- No more than one user per section



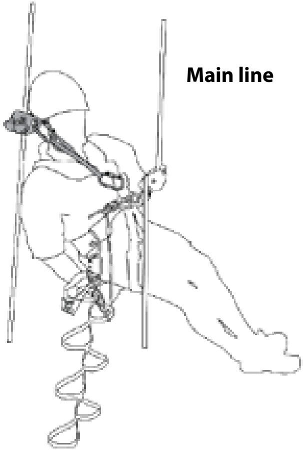
3. Telecommunication Abseiling

The nature of telecommunication structures do not always allow for fall arrest equipment to be used in a safe or effective manner. Therefore, it might be necessary to descend from the top of the tower using rope access to reach the required work area. The following must be adhered to when using rope access on site:

- At least two workers with telecommunication abseiling certificates on site
- Be on two points at all times
- No ascending equipment to be used
- Always rig for rescue

Descending - Moving down on a Rope

1. Install backup device on one of the ropes; this is now your backup line.
2. Install descender on the other rope; this is now your main line.
3. Remove stretch (if there is any) from main line.
4. Put your weight onto the main line
5. Sit down.
6. Grab hold of the lazy rope exiting the descender
7. Ensure your backup remains in a safe fall factor.
8. Press the handle of the descender slowly and descend in a controlled fashion

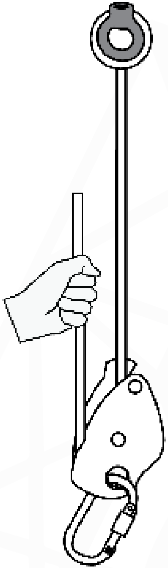


4. Ascending using a Descender

Should a worker descend past the work area or wish to adjust their position, it is recommended to ascend using a descending device instead of ascending equipment in order to facilitate easier rescues.

The steps are as follows:

1. Install handled ascender on main-line
2. Stand up on etrier and release weight from descender
3. Pull the slack rope through descender, causing it to move up on the rope
4. Repeat until desired position is reached
5. Lock descender
6. Remove handled ascender



5. Deviating Work Ropes

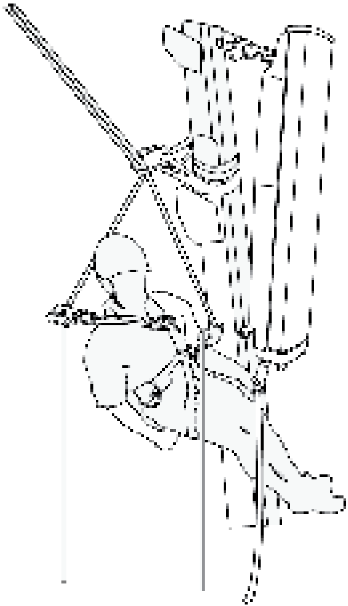
Should a worker desire to reach the sides of a tower and remain there without swinging back or having to keep themselves in position and work with both hands the work ropes should be deviated.

The steps are as follows:

1. Wrap a sling around the tower leg
2. Connect a karabiner in the end of the sling
3. Connect the karabiner on the main line ABOVE the descender
4. Install back-up line in karabiner
5. Lock karabiner
6. Remove when no longer required

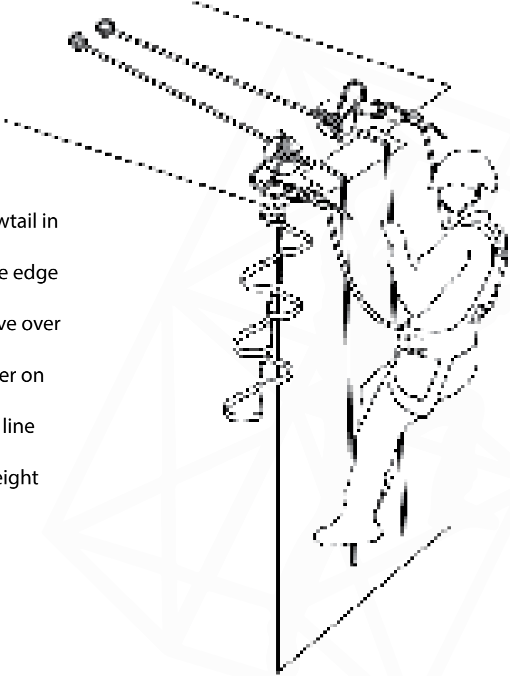
Never connect the karabiner directly to the harness as this will lead to a rescue being impossible.

Never deviate more that 15° from the vertical when using deviations.



6.Passing Over an Edge

When it is necessary to pass over an edge, the following steps are to be followed:



1. When the edge is reached, install the extra cowtail in the knot in the main line.
2. Move the handled ascender and etrier over the edge into the back-up line knot.
3. Remove chest ascender and stand up and move over the edge. When Descending over the edge:
4. When the edge is reached, install the descender on the main line.
5. Install handled ascender and etrier in back-up line knot.
6. Stand up and move over the edge, and put weight onto the descender.
7. Install a backup device on the backup line.
8. Remove handled ascender and etrier.

F. Emergency Procedures

1. Rescues

Ensure that a rescue plan is in place to ensure that a rescue can be performed as quickly and efficiently as possible.

Perform a rescue when;

- A person has fallen and his shock absorbing lanyard has deployed.
- A person has lost consciousness.
- A person is unable to climb down to safety unassisted.

All rescues are different but there are a few things that stay the same throughout:

- Always call the emergency services before attempting a rescue.
- Always establish and maintain communication with the casualty throughout the rescue with either: two-way radios, hand signals or simply talking if the person is within earshot.
- It is recommended that all rescuers are trained first aiders as well.

Ratchet rescue procedure:

1. Call the emergency medical services.
2. Climb up to the casualty.
3. Connect the rope to casualty's top D-link.
4. Connect the suspension trauma sling.
5. Climb above the casualty.
6. Anchor the rescue ratchet with an attachment sling.
7. Remove all the slack between casualty and the descending device.
8. Lift the casualty with the rescue ratchet.
9. Remove the casualty's lanyard.
10. Apply extra friction.
11. Lower the casualty.

Ratchet Rescue

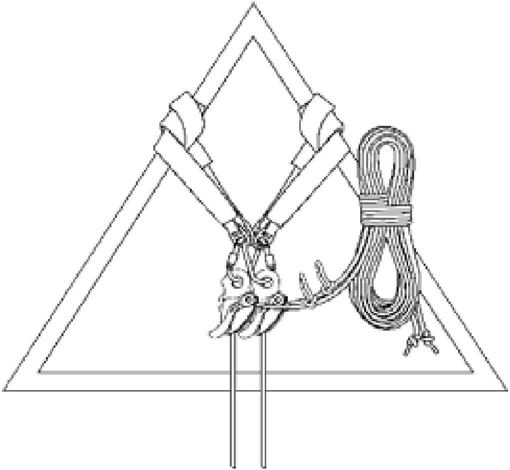
Rescue Kit:

1. Attachment sling
2. Rescue ratchet
3. Descending device
4. Kernmantle rope
5. Suspension trauma sling
6. Attachment karabiner
7. Extra-friction karabiner
8. Kit bag

2. Rigging for Rescue

Rope access work ropes should always be rigged in such a manner that they share the load between at least two independent anchor points. In order to facilitate quick and effective rescue, work ropes should be rigged as follows:

1. Install slings on to separate anchor points
2. Install two descending devices, ensuring that the karabiners of the descenders are connected to both slings
3. Install the work ropes through the descenders ensuring that the amount of slack rope available is long enough to allow for the suspended worker to reach the ground
4. Tie knots in the rope behind the descenders to prevent the rope from slipping through
5. Should a rescue be required, the rescuer is simply to remove the knots, install extra friction and lower the casualty down.



3. Rescue from a Descending Device from Separate Ropes

Should a worker, for any reason, be unable to reach the ground whilst on a descending device, a rescue must be performed. You will need:

- A rescue sling
- 2 x Karabiners
- An extra locker (to be connected to a cowtail):

1. Ascend until you are the same level as the casualty.
2. Connect your extra cowtail to their BOTTOM D-LINK.
3. Connect the rescue sling to YOUR DESCENDER KARABINER and the CASUALTY'S TOP D-LINK.
4. By controlling the casualty's descender,slowly descend the casualty until the descender becomes slack.
5. Remove the casualty's descender from the casualty's harness.

6. Connect the casualty's descender to your harness; ENSURE THAT THE GATES OF BOTH KARABINERS ARE FACING DOWNWARD AND TOWARDS YOU, and remove all the slack and lock it.
7. Remove your backup device.
8. Use the karabiner connecting the casualty's locker sling to their harness and connect this karabiner to the slack ends of BOTH ropes.
9. Whilst keeping tension on the slack ends of the rope running through the extra karabiner, descend down in a controlled fashion.
10. Open both karabiners and remove your BOTTOM D-LINK.

3. Suspension Trauma

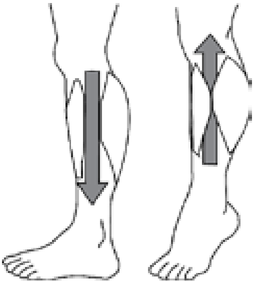
The two biggest risks associated with awaiting a rescue while suspended in a harness, are injury or death due to suspension trauma.

Due to the force of gravity, the blood in the fallen person's body is pulled toward the ground, and pools in the legs.

Due to the constriction of the leg loops of the harness, the blood cannot return to the fallen person's heart. This prevents blood being circulated through a person's body. Thus, less and less blood reaches the brain, leading to loss of consciousness and can lead to brain damage. Finally, this can lead to cardiac arrest and eventually death.

All of this can happen within 15-20 minutes; so it is important to have a good knowledge of the techniques needed to prevent suspension trauma, for yourself, and the casualty that needs to be rescued.

Preventing Suspension Trauma

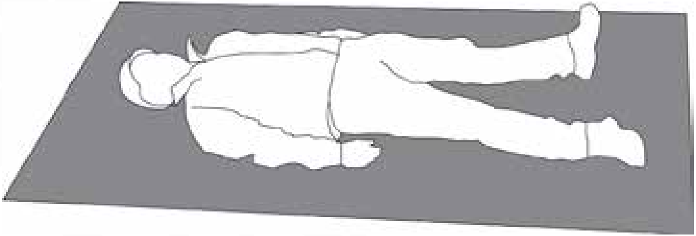


Muscle Pump: Moving your legs, feet and toes will contract the muscles in the legs and force blood back into circulation, which provides the brain with the much needed oxygen to stay conscious.

Suspension Trauma Sling: Will allow you to stand up or to get horizontal, prevent the blood pooling in the legs as well as relieve the pressure of the leg loops which will allow the blood to remain in circulation to the heart and brain. Should the casualty be unconscious, the rescuer can do this for the casualty.

4. Supine Recovery Position

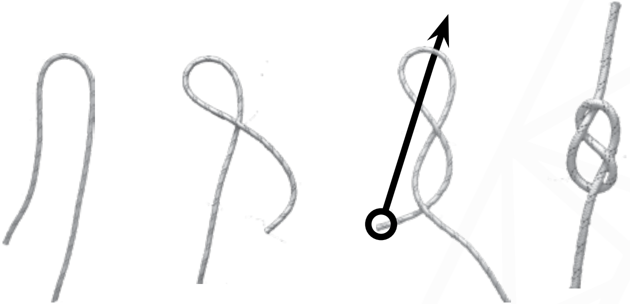
This position allows the blood to flow slowly enough as to prevent Reflow Syndrome, while at the same time keeping the casualty stable and still as to prevent any possible further injury to the casualty's neck and/or spine. This position is also known as the "Supine Position".



G. Knots

- All knots weaken and decrease the length of the rope.
- All knots must be dressed properly; an incorrectly dressed knot will weaken the rope.

1. Figure 8 Stopper Knot

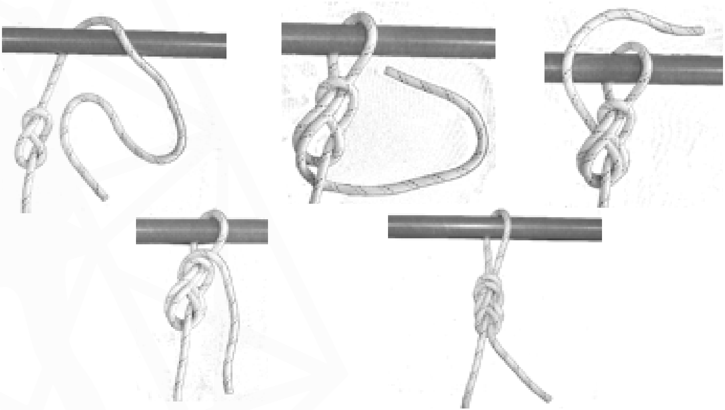


This knot is made in the end of a rope to stop people and equipment from being able of move off the end of the rope.

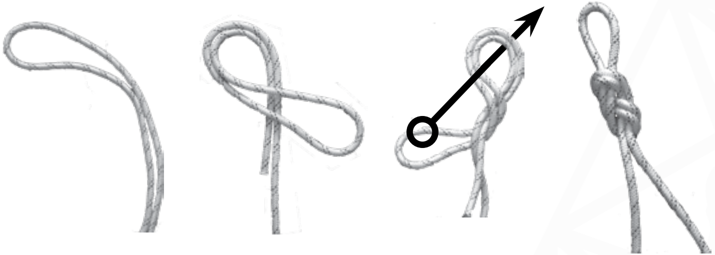
The tail must be longer than your fist, but shorter than your forearm.

2. Figure 8 Follow Through

Simply put, this is a stopper knot that has been completed, which leads to a Figure 8 on a Bight without a connector. This is used to connect cowtails to a harness.

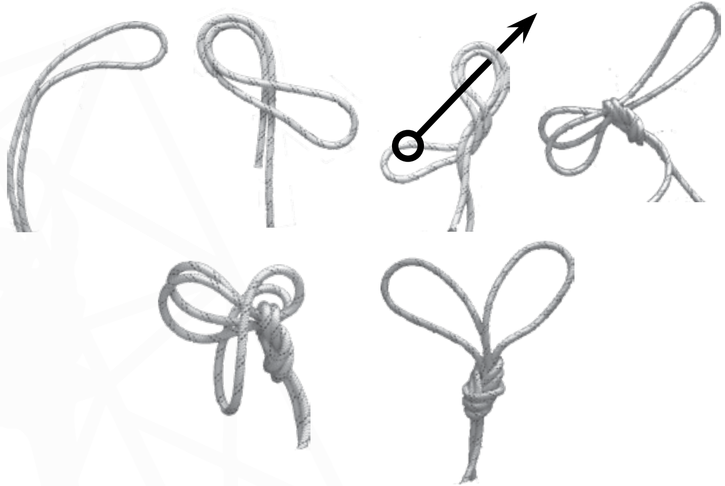


3. Figure of 8 on a Bight



This knot is used to connect the rope to a single anchor point or devices and is used for the installation of a horizontal lifeline as well as during the rescue process.

4. Double Loop Figure 8



This knot is used to connect a rope to two separate anchor points, most commonly used to rig vertical lifelines.

5. Klemheist

Used to create an anchor point on a vertical or diagonal pole. Must be used in a Fall Factor 1 or better.

THE WAY FORWARD

Congratulations on completing Gravity Training's Telecoms Tower Abseiling course.

If found not yet competent (NYC), you will need more training before you can be reassessed. Please do not be discouraged. Not everyone passes on their first attempt. Some learners need a bit more experience to obtain the required knowledge and skills. You will receive comprehensive feedback on your performance and be informed on how to proceed.

At Gravity, we care about the professional development of our clients. Should you wish to take the next step, consider enrolling for some of our other courses. See The Way Forward and the Gravity Skills and Development Map or consult your facilitator/assessor.

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